DAY 13 OF 30

Keeping tables healthy

OPTIMIZE, V-Order and VACUUM.
The housekeeping nobody schedules.

Compaction | V-Order | VACUUM

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THE PROBLEM

Tables get slower without anyone changing them.

1	**Every load**	Adds a few more small files
2	**Every update**	Leaves old files behind
3	**After six months**	Thousands of files per table

The query has to open every one of them. **The data did not grow. The mess did.**

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THE SIMPLE VERSION

Tidying a messy cupboard.

Put things away in a hurry for months and the cupboard fills with half-empty boxes. Finding anything takes longer each week.

Maintenance is **repacking into full boxes and throwing out what is no longer needed**.

✓ **OPTIMIZE**	Merge small files into big ones
✓ **V-Order**	Arrange files for faster reading
✓ **VACUUM**	Remove files no longer in use

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HOW IT WORKS

The three commands

1. **OPTIMIZE** rewrites many small files as fewer large ones
2. **V-Order** sorts and compresses Parquet for fast reads
3. **VACUUM** deletes old files past the retention period
4. The **Maintenance** dialog on a lakehouse table runs all three

Default retention is 7 days

VACUUM keeps a week of history. Shorter than that and time travel stops working for older versions.

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FOR ENGINEERS

What changed, and what to check

V-Order is off by default in new workspaces

The default profile favours fast writes. Reports read faster with V-Order, so turn it on for serving tables.

Resource profiles set the defaults

writeHeavy is the default. The read-heavy profiles enable V-Order and optimised writes for you.

Auto compaction is opt-in

Microsoft recommends it for ingestion tables. It compacts after writes so you do not have to schedule it.

Order matters

Compact first, then vacuum. Vacuuming first just removes files you were about to rewrite.

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THE HONEST PART

How often is enough?

MAINTAIN OFTEN

Tables loaded many times a day

Tables that Power BI reads directly

Tables with frequent updates or deletes

LEAVE ALONE

Large tables that are rarely written

Bronze tables nobody queries

Tables rebuilt in full each run

> **The catch:** maintenance uses capacity too. OPTIMIZE on a huge table at 9 a.m. competes with everyone's reports. Schedule it like any other heavy job.

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FROM REAL WORK

Maintenance on a real platform

A Databricks platform for a financial services company, with daily loads into Delta Lake.

1. **Partitioning** on the large Delta tables
2. **Compaction** to merge small files into larger ones
3. **Z-Ordering** to keep related rows close together
4. **Query scan times** improved as a result

The same three ideas apply to **every Delta table in Fabric**.

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SAVE THIS

Maintenance in five lines

1. **Small files** are the silent slowdown
2. **OPTIMIZE** compacts them
3. **V-Order** speeds up reads, check it is on
4. **VACUUM** cleans up, mind the 7 days
5. **Schedule it** away from peak hours

Tomorrow: The on-premises data gateway.
Reaching data that sits inside a factory or office.

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